GEO-HD-1

Higher-Dimensional Generative Geometry of the UNNS Substrate

Exact Arithmetic Geometry, Fibre Transport, Generative Histories, and Higher-Dimensional Obstructions

Milestone designation: GEO-HD-1
Date: 30 October 2026
Document class: Mathematical foundations and first-experiment specification
Status: Active, new investigation; not a frozen GEO-I release
Reference baseline: GEO-I v001–v012; completed Papers A and B; *The Mathematical Foundations of the UNNS Substrate*, Volumes I and II.

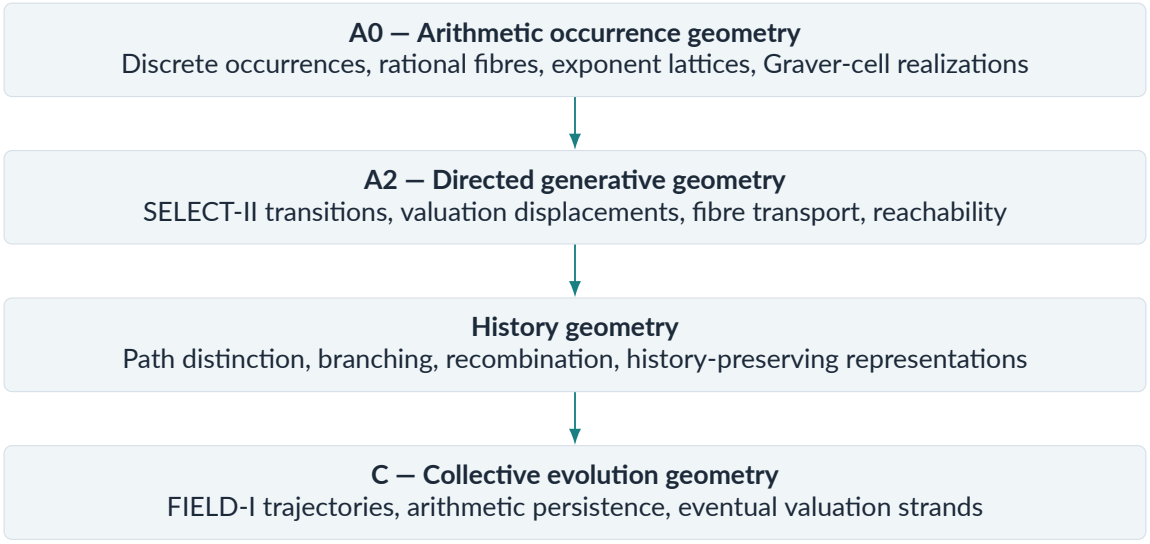
The central goal is to establish new arithmetic-specific theorems connecting occurrence fibres, labelled generation, histories, and collective evolution. Illustrative geometry is subordinate to exact definitions and proofs.

Executive research mandate

The GEO-I programme established laws governing the geometry of individual numerical occurrence fibres. GEO-HD-1 investigates the *geometry of generation itself*: how arithmetic structures transform, branch, merge, preserve information, and become obstructed under generative evolution. Its aim is original theorem-level mathematics rather than a third synthesis paper, a rendering, or a relabelling of existing results.

Governing question. Can the arithmetic structure of UNNS determine how occurrence geometries transform, branch, merge, preserve information, or become obstructed under generative evolution—and can these relationships be expressed as rigorous laws in three, four, and arbitrary dimensions?

Four distinct layers



Scope restriction. The connecting maps must be explicitly defined. Neither the arithmetic occurrence rule nor SELECT-II alone determines the FIELD-I collective operator. Cell incidence is likewise not identical to dynamic graph adjacency. No Euclidean metric or physically meaningful fourth coordinate follows merely from drawing axes.

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1 Exact mathematical foundation

1.1 Arithmetic occurrence space

Define the discrete occurrence space

$$\mathcal{O} = \left\{ (N, M, x) : N, M \in \mathbb{N}_{>0}, \quad x = \frac{M(N+1)^2}{N} \right\}. \tag{1}$$

For a fixed reduced positive rational $x = a/b$, write $\mathcal{O}_x$ for its horizontal section. Define the maximal square-root part of the numerator by

$$s(a) = \prod_p p^{\lfloor v_p(a)/2 \rfloor}. \tag{2}$$

The associated exact exponent-coordinate fibre is

$$\begin{aligned} E(x) = \{ (\alpha_p)_{p|s(a)} : 0 \le \alpha_p \le v_p(s(a)), \\ d = \prod_p p^{\alpha_p} \ge 2, \quad d \equiv 1 \pmod{b} \}. \end{aligned} \tag{3}$$

The map from an occurrence to its exponent coordinates is

$$(N, M, x) \longmapsto (v_p(N+1))_{p|s(a)}, \tag{4}$$

with inverse, for $d = \prod_p p^{\alpha_p}$,

$$N = d - 1, \qquad M = \frac{a(d-1)}{bd^2}. \tag{5}$$

The map is a bijection by Volume I, Theorem 1.2. Hence the arithmetic section and the exponent-coordinate fibre encode the same occurrences, although their Euclidean/cellular realizations are additional modelling choices.

1.2 Full valuation geometry

For positive integer states use the finite-support valuation lattice

$$\mathcal{V}_{\mathbb{N}} = \bigoplus_{p \text{ prime}} \mathbb{Z}_{\ge 0}. \tag{6}$$

The SELECT-II graph has labelled edges

$$m \xrightarrow{N} T_N(m) = \frac{m}{N}(N+1)^2, \qquad N \mid m. \tag{7}$$

The exact arithmetic displacement is

$$\boxed{\mathbf{v}(T_N(m)) - \mathbf{v}(m) = 2\mathbf{v}(N+1) - \mathbf{v}(N).} \tag{8}$$

This is an identity in full finite-support valuation space, not a Euclidean motion or cost function. Three- and four-coordinate pictures are coordinate projections and may erase information. For rational FIELD-I states the corresponding full valuation lattice allows negative exponents: $\bigoplus_p \mathbb{Z}$.

1.3 History-unfolded geometry

For a chosen initial state m_0, define an admissible finite history

$$h = (N_1, \dots, N_\ell), \qquad N_j \mid m_{j-1}, \quad m_j = T_{N_j}(m_{j-1}). \tag{9}$$

The history space $\mathcal{H}(m_0)$ is ordered by prefix extension and has the endpoint map

$$\varepsilon : \mathcal{H}(m_0) \to \mathbb{N}_{>0}, \qquad h \mapsto m_\ell. \tag{10}$$

This distinguishes paths even when the endpoint map is not injective. A fourth time coordinate alone does not retain the entire path. A path unfolding, edge-label sequence, or equivalent history-identifying structure is needed.

2 First theorem target: exact fibre transport

The first substantive bridge is between the static occurrence fibre and a labelled SELECT-II transition. Define, for each positive integer m,

$$s(m) = \prod_p p^{\lfloor v_p(m)/2 \rfloor}, \qquad D(m) = \{d \geq 2 : d \mid s(m)\}. \tag{11}$$

Proposition 2.1 (GEO-HD-T1: fibre-preservation criterion)**.** For a SELECT-II edge $m \xrightarrow{N} t$ with $t = m(N+1)^2/N$,

$$\boxed{D(m) \subseteq D(t) \iff \forall p \mid N : \ v_p(N) = 1 \text{ and } v_p(m) \text{ is odd.}} \tag{12}$$

The condition is vacuous for $N = 1$.

Proof. Fibre inclusion is equivalent to $s(m) \mid s(t)$. For every prime $p \mid N$, coprimality of N and $N+1$ yields $v_p(t) = v_p(m) - v_p(N)$. Thus the exponent of p in $s(t)$ is at least the exponent in $s(m)$ precisely when

$$\left\lfloor \frac{v_p(m) - v_p(N)}{2} \right\rfloor = \left\lfloor \frac{v_p(m)}{2} \right\rfloor. \tag{13}$$

Since $v_p(N) \geq 1$, this is possible exactly when $v_p(N) = 1$ and $v_p(m)$ is odd. For primes not dividing N, the exponent of the state remains unchanged or increases through the factor $(N+1)^2$; none of those primes can cause a loss. These conditions are therefore necessary and sufficient. □

2.1 Quantitative transport: retained, lost, gained

Put $g = \gcd(s(m), s(t))$. Counting nontrivial divisors gives exact transport statistics:

$$\text{Retained}(m,t) = \tau(g) - 1, \tag{14}$$
$$\text{Lost}(m,t) = \tau(s(m)) - \tau(g), \tag{15}$$
$$\text{Gained}(m,t) = \tau(s(t)) - \tau(g), \tag{16}$$

where τ is the positive divisor-counting function.

This distinguishes preserving the *number* of occurrences from preserving the *occurrence indices* themselves.

2.2 Exact benchmark: transitions from 72

The initial fibre is $D(72) = \{2, 3, 6\}$. All rows follow the original SELECT-II grammar and exact prime factorization.

SELECT-II edge	**Resulting fibre** $D(t)$	**Lost indices**
$72 \xrightarrow{1} 288$	$\{2, 3, 4, 6, 12\}$	None
$72 \xrightarrow{2} 324$	$\{2, 3, 6, 9, 18\}$	None
$72 \xrightarrow{3} 384$	$\{2, 4, 8\}$	$\{3, 6\}$
$72 \xrightarrow{6} 588$	$\{2, 7, 14\}$	$\{3, 6\}$

The edge $72 \to 384$ preserves multiplicity $R = 3$ but replaces two of the three index routes. In particular,

$$R(t) = R(m) \quad \not\Longrightarrow \quad D(t) = D(m). \tag{17}$$

The FIELD-I collective map, by contrast, preserves all existing fibre indices along its own orbit (Volume II, Theorem 1.3). Thus the two declared continuation grammars have sharply distinguishable structural effects.

PROOF STATUS: DERIVED IN THIS SPECIFICATION

GEO-HD-T1 and the counting identities follow by exact divisor arithmetic. Their originality relative to the complete frozen registry and the wider mathematical literature remains a separate audit question; do not claim established priority.

3 Exact higher-dimensional obstruction

For distinct primes $p_1, \dots, p_k$, choose

$$x_k = (p_1 \cdots p_k)^2, \qquad k \geq 2. \tag{18}$$

The genuine integer occurrence fibre in prime exponent coordinates is

$$E(x_k) = \{0, 1\}^k \setminus \{\mathbf{0}\}. \tag{19}$$

Under the expressly chosen *coordinate-cubical grammar*, include a coordinate face only if all its vertices belong to the fibre. Its union of admissible cells and its convex hull are, respectively,

$$U_k = \{u \in [0, 1]^k : \max_i u_i = 1\}, \tag{20}$$

$$H_k = \operatorname{conv}(E(x_k)) = \{u \in [0, 1]^k : \textstyle\sum_i u_i \geq 1\}. \tag{21}$$

$$\boxed{U_k \subsetneq H_k \qquad \text{for every } k \geq 2.} \tag{22}$$

Certificate. The point $(1/2, \dots, 1/2)$ lies in H_k but not in U_k.

For $k = 3$ and $(p_1, p_2, p_3) = (2, 3, 5)$, this is the exact fibre of $x_3 = 900$: seven cube vertices, with the origin excluded.



Cell union U_3
Three outer square faces; the origin and interior are absent

Convex hull H_3
Additional truncation face and interior points

Figure 1. Schematic projection of the $k = 3$ coordinate-cell model. The equations and point-membership certificate, rather than this rendering, establish noncoverage.

The result is *grammar-relative*. It does not contradict Paper A's planar stabilization theorem, whose Graver-cell construction and enlarged-box hypotheses differ. The next task is to determine whether the GEO-I Graver grammar admits a similar obstruction or whether additional admissible cells restore coverage.

4 Exact benchmark requirements

State	Mathematical purpose	Required exact test
6	Full valuation-space transition geometry	Four labelled outputs $24, 27, 32, 49$; faithful versus projected coordinates
72	Multiplicity, fibre transport, histories	Three occurrences; two different histories to 72; preserved/lost indices
900	Genuine 3-coordinate fibre geometry	Seven exponent vertices; exact hull/cell comparison; topology/embedding checks
General k	Higher-dimensional obstruction	Prove or refute coverage and topology extensions under each explicitly declared cell grammar

4.1 Benchmark 6: complete local valuation data

The four SELECT-II transitions from 6 are

$$6 \xrightarrow{1} 24, \quad 6 \xrightarrow{2} 27, \quad 6 \xrightarrow{3} 32, \quad 6 \xrightarrow{6} 49. \tag{23}$$

In the projection (v_2, v_3, v_7), source 6 has coordinates $(1, 1, 0)$ and the four destinations have coordinates $(3, 1, 0)$, $(0, 3, 0)$, $(5, 0, 0)$, $(0, 0, 2)$. These projected vectors are computed from each complete finite-support valuation vector. Other three-coordinate projections may identify distinct integer states.

4.2 Benchmark 72: occurrence points and merging histories

The occurrence surface has exactly the three points

$$(N, M, x) = (1, 18, 72), \quad (2, 16, 72), \quad (5, 10, 72). \tag{24}$$

The following two SELECT-II histories are distinct even though their initial state, endpoint and path lengths are identical:

$$\begin{aligned} 1 &\xrightarrow{1} 4 \xrightarrow{1} 16 \xrightarrow{2} 72, \\ 1 &\xrightarrow{1} 4 \xrightarrow{2} 18 \xrightarrow{1} 72. \end{aligned} \tag{25}$$

The history-unfolded space distinguishes them; their endpoint/time projections do not.

4.3 FIELD-I scaling: scoped valuation strands

For the explicitly adopted collective map $C(x) = x + L(x)^2/G(x)$, Volume II proves eventual exact scaling and single-prime rigidity. Under the precise conditions yielding

$$x_{n+P} = p^{2s} x_n \quad (n \geq n_0), \tag{26}$$

prime valuation immediately gives

$$\mathbf{v}(x_{n+P}) = \mathbf{v}(x_n) + 2s\,\mathbf{e}_p. \tag{27}$$

This inherited implication yields at most P phase subsequences of translated valuation rays after entry into the stipulated final regime. The new questions concern their intersections, transient-to-strand transition, and intrinsic arithmetic invariants. The illustrative regime-I orbit $72 \to 108 \to 144 \to 288 \to \cdots$ must remain distinct from SELECT-II edges out of 72; for regime I, eventual three-step scaling is addressed in Volume II, Theorem 2.4.

5 Research execution and acceptance gate

HD-1A. Formalization. Fix occurrence, valuation, cell, route and history categories; build a source crosswalk against GEO-I and the SELECT/FIELD records. Specify domain restrictions, admissibility, and which maps are faithful versus projected.

HD-1B. Fibre transport. Develop GEO-HD-T1 into a classification of index gain/loss. Define cross-fibre geometric maps and test preservation of cells, hulls and topological invariants.

HD-1C. Genuine 3D and 4D topology. Investigate convex coverage, face-to-face compatibility, H_1, H_2, cavities, and minimal counterexamples. Compare the coordinate-cell and frozen GEO-I Graver-cell grammars without treating them as interchangeable.

HD-1D. Generative histories. Classify branching, recombination, projection collisions, and common futures through exact arithmetic. Generic path unfolding alone does not satisfy the discovery threshold.

HD-1E. Collective evolution. Translate valid FIELD-I theorems into valuation-strand geometry. Investigate invariants governing the transition to persistence and distinguish phase trajectories.

5.1 Reproducible working package

Use a separate provisional workspace such as `UNNS_GEO_HD/`. The package should include definitions, a source crosswalk, theorem and open-problem registries, exact benchmark data for 6, 72 and 900, deterministic integer/rational verifiers, and diagrams generated from verified tables. Numerical topology must be checked against explicit combinatorial incidence. No local or GitHub reference release is to be modified without separate authorization.

Acceptance gate. GEO-HD-1 succeeds when (i) the arithmetic/geometric bridge is proved, (ii) projection and history information loss are formally characterized, and (iii) at least one substantive higher-dimensional or generative-transport theorem or sharp obstruction is established with exact reproducible certificates and delimited hypotheses. New results must be compared with the frozen sources and relevant existing mathematical theory before asserting priority.

6 Strategic outcome

GEO-I established laws about *what geometry numerical occurrences possess*. GEO-HD-1 asks the more fundamental dynamical question:

> *How does that geometry change when the UNNS Substrate generates a new state, and which aspects of the original structure survive the transformation?*

The fibre-transport criterion and coordinate-cubical obstruction provide two initial exact propositions to strengthen and generalize. They are not by themselves completion of the milestone. The priority is arithmetic-specific discoveries before any Paper C or public synthesis.

Source and scope crosswalk

- **GEO-I v001–v012 and Papers A/B** (frozen): occurrence-fibre geometry, planar coverage under specified Graver cells, and rank-three collisions in a specified binary relation-lattice model. No automatic transfer of those statements to other cell grammars.
- **Mathematical Foundations, Volume I** (23 October 2026): Theorem 1.2 (rational fibre), Theorems 5.2–5.5 (SELECT-II graph, predecessor/label relations), Chapter 6 (histories/reachability). Scope A0 and A2 as stated there.
- **Mathematical Foundations, Volume II** (23 October 2026): Theorem 1.3 (monotone fibres under C), Theorem 2.4 (regime-I scaling), and Chapter 4–5 scaling and single-prime results. Scope C: the explicitly adopted collective map, not the nest arithmetic alone.
- **GEO-HD handoff, 30 October 2026**: `CTX_GEO_HD_NEXT_MILESTONE_20261030.md`; defines the separated-workspace policy, exact benchmarks, and theorem-first acceptance conditions.

Document boundary: this is a milestone specification with two derived propositions, not an archived theorem release, a formal proof-assistant certificate, or an assertion of physical interpretation.